SDV Resource Governor -- Technical Pitch Deck (8 Slides)

Slide 1: Executive Overview -- Deterministic Physics First

- **Mission**: Govern admission, placement, live arbitration, and graceful degradation across **8 coupled physical resource classes** (‘power‘, ‘cooling‘, ‘positions‘, ‘mast‘, ‘mass_stability‘, ‘data‘, ‘compute‘, ‘time‘) on a Software-Defined Vessel.
- **Core Principle**: Hard physical and life/vehicle safety limits are satisfied **first**; mission capability score, free margin, and displacement cost are optimized **second**. No priority weight can ever override a hard physical limit or drone ditching deadline.
- **Zero Opaque Runtime AI**: 100% deterministic, inspectable math (exact cooling curve, integer Watts/kg/bps, parameterized witness explanations, 1 Hz internal simulation stepping).

Slide 2: Fail-Closed Validation & Defeating Legacy Defects

- **Strict Schema & Type Discipline (‘B2‘, ‘F3‘, ‘H06‘, ‘H13‘, ‘R2‘, ‘R4‘)**:
  - Rejects ‘NaN‘, ‘Infinity‘, negative or zero budgets, booleans used as numbers (‘True != 1‘), string booleans (‘"false"‘), unknown YAML/HTTP JSON fields (‘UNKNOWN_FIELD‘), dependency cycles, and invalid cooling coupling (‘cooling_water_w + cooling_air_mast_w == sustained_power_w‘).
  - Reproduces and defeats all **10 adversarial input flaws** (‘resource_engine_repro.json‘) and all **25 verifier mutation classes** (‘false_accepts = 0‘).
- **Six Declarative Guest Cards (‘G1‘-‘G6‘) with Field-Level Provenance**:
  - Official ‘G1‘ (‘45 kg‘, ‘1.5 kW‘ air), ‘G2‘ (‘3,500 kg‘, ‘20/6 kW‘), ‘G3‘ (‘5,500 kg‘, ‘25/1 kW‘), ‘G4‘ (‘140+850 = 990 kg‘ per errata, ‘35/10/2 kW‘), ‘G5‘ (‘3,000+25 = 3,025 kg‘, ‘45/8 kW‘), plus custom civil card ‘G6‘ (‘420 kg‘, ‘port_only=true‘, all fields ‘ASSUMED‘).

Slide 3: Scenario A (D1) -- 8-Class Admission & Errata Proof

- **Organizer Errata Verified (‘DAT-02‘, ‘DAT-03‘)**:
  - With ‘G4 = 990 kg‘, exactly **3 of 5** four-guest combinations exceed ‘10,000 kg‘: ‘{G1,G2,G3,G4}=10,035 kg‘, ‘{G1,G2,G3,G5}=12,070 kg‘, ‘{G2,G3,G4,G5}=13,015 kg‘.
  - The 2 mass-feasible sets ‘{G1,G2,G4,G5}=7,560 kg‘ and ‘{G1,G3,G4,G5}=9,560 kg‘ prove that **passing mass does not imply passing placement**: ‘G5‘’s antenna collides with ‘G1‘ on ‘mast_3‘ (‘1.5+1.2=2.7 kW > 2.0 kW‘ air) and with ‘G4‘ drone clearance on ‘hangar_roof‘!
- **Single Platform Modification Counterfactual (‘R6.1‘)**:
  - Adding a second guest mast slot ‘mast_4‘ (‘<= 60 kg‘, ‘<= 2.0 kW‘ air) resolves the ‘G1‘/‘G5‘ mast collision and upgrades both ‘{G1,G2,G4,G5}‘ and ‘{G1,G3,G4,G5}‘ to unconditional **‘YES‘** across the 14-day Mission Envelope A.10.

Slide 4: Scenario B (D3) -- Live Arbitration & Causal Acoustic Handshake

- **The Conflict**: ‘G3‘ AUV is executing a ‘40 min‘ quiet acoustic survey (‘SURVEY_QUIET‘, ‘1 kW‘, ‘3 kn‘) when ‘G4‘ UAV reports **‘18%‘ battery (‘< 20%‘ emergency -> ‘12 min / 720 s‘ max endurance)** requiring a turn into ‘22 kn‘ wind and ‘35 kW‘ recovery/charging power.
- **Causal Command-ACK & Precondition Verification (‘R1.1‘, ‘R1.2‘)**:
  - D3 approval at ‘T=480 s‘ immediately registers a firm ‘35 kW‘ reservation (‘D3_PLAN_V1_G4_RECOVERY‘) for ‘G4‘ recovery/charging.
  - Turning into the wind requires a verified ‘ACOUSTIC_HOLD_CONFIRMED‘ ACK from ‘guest_id=’G3’‘ matching the active ‘command_id‘ and ‘plan_version‘, plus ‘power_reserve_w >= 35 kW‘ and ‘deck_clear=True‘.
  - **‘rescue_primary‘**: Issues ‘ACOUSTIC_HOLD‘ at ‘T=480 s‘, receives ACK at ‘T=495 s‘, turns into wind, and lands ‘G4‘ at ‘T=855 s‘ (‘345 s‘ margin before ‘T=1,200 s‘ deadline).
  - **‘survey_primary‘**: Completes ‘240 s‘ of survey pass, issues ‘ACOUSTIC_HOLD‘ at ‘T=720 s‘, receives ACK at ‘T=735 s‘, turns into wind, and lands ‘G4‘ at ‘T=1,095 s‘ (‘105 s‘ margin before ‘T=1,200 s‘ deadline).

Slide 5: Scenario B (D4) -- Graceful Degradation & Zero Infeasible Ticks

- **Honest Power vs. Thermal Accounting (‘DEG-01‘..‘05‘, ‘R1.1‘)**:
  - When generator reserve drops from ‘200 kW‘ to ‘100 kW‘ at ‘T=720 s‘, firm committed power including ‘G3‘’s ‘25 kW‘ turnaround recharge reservation and ‘G4‘’s ‘35 kW‘ D3 recovery reservation is ‘25 + 35 + 45 = 105 kW > 100 kW‘ (‘5 kW‘ firm deficit).
  - Under both ‘survey_primary‘ and ‘rescue_primary‘, D4 immediately steps down **‘G5‘ (‘45 kW -> 8 kW‘ ‘DEGRADED_70PCT_LOSS‘)** at ‘T=720 s‘, reducing committed power to ‘68 kW <= 100 kW‘ (‘32 kW‘ free margin) and guaranteeing **0 infeasible ticks (‘91/91‘ feasible)** across the entire mission!
  - At ‘30 deg C‘ seawater with ‘-40%‘ cooling loop loss (‘72 kW‘ thermal capacity vs ‘100 kW‘ power), **chilled-water cooling binds first** (‘79.8 kW > 72 kW‘) and steps down ‘G5‘ (‘79.8 kW -> 43.5 kW‘).

Slide 6: Scenario C -- Crew Regimes (14 / 6 / 0) & Resilience

- **Crew Regime & Deck Clearance Enforcement (‘CRW-01‘..‘03‘, ‘R3‘, ‘R6.2‘)**:
  - **14 Crew**: ‘9‘ deck crew -> ‘G3‘ (‘4‘ crew) and ‘G4‘ (‘3‘ crew) at-sea vehicle replacement permitted in A’ window (‘4-8 kn‘, ‘wind <= 20 kn‘, ‘SS <= 3‘). Simultaneous ‘G2‘ winch (‘0-5 kn‘) + ‘G4‘ flight ops (‘5-12 kn‘ into wind) remain forbidden by Annex A.5 ‘deck_clear‘ and disjoint speed windows.
  - **6 Crew**: ‘2‘ deck crew -> serialized manual launch/recovery (‘G2‘ or ‘G4‘) permitted with ‘60-180 s‘ watch handover; at-sea vehicle replacement (‘min_crew >= 3‘) blocked.
  - **0 Crew**: Zero physical installations or manual deck evolutions at sea (‘CRW-02‘); ‘G6‘ (‘420 kg > 250 kg‘, bay install) is ‘port_only=True‘ across all regimes.

Slide 7: Dual-Platform Architecture & Live Ventuno Q Parity

- **Decoupled Laptop + Edge Board Execution (‘B1‘, ‘DEP-01‘..‘03‘)**:
  - **MacBook Pro (‘arm64‘, Python 3.13.15)**: Full test suite, 1 Hz event-driven simulator, 90 s (20x) compressed demo, and stateful 2D operator console (‘http://127.0.0.1:8765‘).
  - **Arduino Ventuno Q (‘training06‘, Qualcomm Dragonwing IQ8 ‘aarch64‘, 16 GB RAM, Ubuntu 24.04.4)**: Live SSH deployment at ‘/home/arduino/sdv-resource-governor‘.
  - **Bit-for-Bit Semantic Parity**: Identical ‘campaign_semantic_sha256‘ verified across ‘arm64‘ macOS and live ‘aarch64‘ Ventuno Q (‘training06‘).

Slide 8: Single-Screen 2D Operator Console & Fail-Closed Evidence

- **Stateful Single-Screen 2D Operator Console (‘B10‘, ‘H12‘, ‘R2‘, ‘R5‘)**:
  - Persistent ‘GovernorSession‘ (‘session_catalog‘, ‘manifest_registry‘, ‘snapshot_id‘, ‘state_version‘), per-tick 8-class budget and 2D vessel topology synchronization, 90 s (‘1000 ms/tick = 20x‘) Scenario B playback, perturbation injector, and live Scenario A guest card editor.
- **Fail-Closed Evidence Verifier (‘B8‘, ‘H05‘, ‘R4‘)**:
  - Recomputes all summary metrics and semantic hashes from raw JSON logs, verifies all 49 acceptance criteria ‘<testcase>‘ nodes in ‘tests.*‘ with portable ‘requirements_matrix.csv‘, cross-checks ‘git_commit‘ and ‘environment‘, enforces ‘<= 200 ms‘ tick deadlines, and rejects **25 / 25** adversarial mutations (‘false_accepts = 0‘).